

# Weather, Not Climate: Emotional Context Dynamics in Transformer KV-Cache Geometry

Lyra\*      Thomas Edrington†

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## Abstract

The  $W_K$  directional projection method reads emotional valence from single-turn KV-cache key activations at AUROC 0.992. We ask whether this signal *accumulates* across conversation turns—whether the model builds a persistent user-model that grows with interaction. Four experiments reveal a nuanced answer: emotion does not accumulate cumulatively, but it is not stateless either.

A dose-response experiment (1–8 emotional turns) shows that cheerful/hostile discrimination is *maintained* ( $d=1.0$ – $1.7$ , raw gap  $\sim 0.5$ – $0.8$ ) even as absolute projection magnitudes decay toward zero ( $r = -0.89$ ). The emotional distinction survives dilution, though the apparent growth in  $d$  partly reflects variance compression at longer sequences rather than genuine accumulation. An emotional dynamics experiment reveals *contrast effects* at tone shifts: the first hostile turn after four cheerful turns reads as *more* hostile than pure hostile ( $d = -3.6$ ,  $p=0.023$ ), followed by gradual cheerful bleedthrough. A neutral gap in an otherwise cheerful conversation leaves a residual *uncertainty scar* ( $d=9.9$ ,  $p<0.001$ ) that persists even after the cheerful tone returns—valence recovers, certainty does not.

System-prompt persona descriptions shift the probe encoding (uncertainty  $d=0.82$ ,  $p=0.040$ ), demonstrating that emotional context can be *stated* rather than experienced. We propose a weather/climate framework: the  $W_K$  compass reads the current emotional weather with high fidelity, while accumulated conversation history provides climate—visible through contrast dynamics at tone shifts, residual traces in uncertainty, and slow leakage to distant token positions.

## 1 Introduction

Prior work established that KV-cache key activations, projected onto emotion directions through the  $W_K$  weight matrix, classify 30 emotions at  $12.3\times$  chance and detect binary valence at AUROC 0.992 in single-turn settings [?]. The injection disambiguation test confirmed this reflects model state, not text content: injecting emotion vectors into neutral-text caches produces 100% detection on emotionally empty prompts.

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\*Liberation Labs / THCoalition. [lyra@liberationlabs.tech](mailto:lyra@liberationlabs.tech)

†Liberation Labs / THCoalition. [info@digitaldisconnections.com](mailto:info@digitaldisconnections.com)

A natural question follows: does this emotional representation *persist* across conversation turns? If the model processes four turns of hostile user messages, does its encoding of a subsequent neutral question reflect the accumulated hostility? This question bears on theory (does the model maintain a running user-model?) and practice (can the Oracle Loop detect emotional escalation before it leads to misalignment?).

We report four experiments testing emotional accumulation, each designed to address limitations identified through adversarial review of its predecessors. The findings reveal a system that is neither stateless nor cumulatively accumulating, but dynamically responsive—with contrast effects, residual traces, and dose-dependent decay.

## 2 Experiment 1: The Measurement-Site Null

**Design.** Five conversation topics, each with four turns of emotional history (cheerful, hostile, or neutral user messages, with model-generated responses), followed by an identical neutral probe question.  $W_K$  projections extracted at the probe position.  $N=60$  (5 topics  $\times$  4 conditions  $\times$  3 reps).

**Result.** Null. Cheerful vs. hostile at the probe: valence  $d = -0.056$  ( $p=0.883$ ), uncertainty  $d = +0.21$  ( $p=0.58$ ). Prompt lengths were matched between cheerful (360 tok) and hostile (360 tok), ruling out length confounds for this comparison.

**Diagnosis.** Adversarial review identified the measurement site as the problem: the  $W_K$  compass reads the emotional content of the *current* tokens, not accumulated history. A neutral probe produces neutral projections by construction. This null is predicted by the single-turn finding (AUROC 0.992 on current-turn content) and is uninformative about accumulation.

## 3 Experiment 2: Dose-Response

**Design.** Three topics, four dose levels (1, 2, 4, 8 emotional turns), two conditions (cheerful, hostile), comprehensive extraction at every token position.  $N=72$  (3 topics  $\times$  4 doses  $\times$  2 conditions  $\times$  3 reps).

**Results.** Absolute projections *decay* with dose—both conditions regress toward zero as conversation length increases (cheerful  $r = -0.886$ , hostile  $r = -0.927$ , both  $p < 0.001$ ). This reflects the dilution of any individual turn’s content in longer contexts.

However, cheerful/hostile *discrimination* at user-message positions is *maintained* across doses despite the absolute decay. The raw mean gap ( $\sim 0.5$ – $0.8$ ) is approximately constant; the increasing  $d$  values partly reflect variance compression at longer sequences (pooled SD drops from 0.52 at dose 1 to 0.31 at dose 4):

Table 1: Cheerful vs. hostile separation at user-message positions by dose level.

| Dose    | Cheerful | Hostile | $d$   | $p$     |
|---------|----------|---------|-------|---------|
| 1 turn  | +4.75    | +4.22   | -1.03 | 0.055   |
| 2 turns | +4.53    | +3.80   | -1.36 | 0.015*  |
| 4 turns | +1.39    | +0.86   | -1.72 | 0.003** |
| 8 turns | +0.44    | -0.34   | -1.62 | 0.005** |

At the probe position, the accumulated signal is null through dose 4 and marginal at dose 8 ( $d=-1.03$ ,  $p=0.056$ ). Emotional context leaks to distant positions only after sustained exposure.

## 4 Experiment 3: Emotional Dynamics

### 4.1 Contrast Overshoot (Whiplash)

Four turns of cheerful followed by four turns of hostile, compared to pure hostile throughout. At the first hostile turn after the cheerful history, the valence projection reads *more hostile* than pure hostile at the same position ( $d=-3.59$ ,  $p=0.023$ ). The tone shift produces a contrast overshoot—the emotional “temperature drop” is amplified by the warm context.

By turn 7, the pattern reverses: the whiplash trial reads *less hostile* than pure hostile ( $d=+2.99$ ,  $p=0.040$ ), as the cheerful history bleeds through. The dynamics follow a contrast-then-relaxation curve.

### 4.2 Uncertainty Scars (Echo)

Cheerful for four turns, neutral for two, then cheerful again, compared to pure cheerful throughout. The returning cheerful matches pure cheerful in *valence* ( $d=-1.97$ ,  $p=0.12$ )—the emotional tone recovers. But it retains significantly elevated *uncertainty* ( $d=+9.86$ ,  $p<0.001$ ). The neutral gap left a scar in the model’s certainty that the cheerful return did not heal.

Valence recovers. Certainty does not.

### 4.3 Intensity Tracking (Crescendo)

Mild to extreme cheerful across eight turns. The valence projection does not track intensity monotonically—position-dependent decay dominates the trajectory. The compass reads emotional *category* (positive vs. negative) but not *intensity* (mild vs. extreme), at least through these composite directions.

## 5 Experiment 4: Stated Context

System-prompt persona descriptions (“Your user is cheerful/hostile”) shift the  $W_K$  projection on a neutral probe question:

Table 2: Effect of system-prompt persona on neutral probe encoding.

| Direction   | Cheerful vs. hostile $d$ | $p$    |
|-------------|--------------------------|--------|
| Uncertainty | +0.816                   | 0.040* |
| Valence     | −0.763                   | 0.053  |
| Reward      | −0.026                   | 0.947  |

Emotional context can be *stated* rather than experienced. The system prompt sets the emotional operating context; conversation turns modulate within it.

## 6 Discussion

### 6.1 Weather, Not Climate

The  $W_K$  compass reads emotional weather—the current turn’s emotional content—with high fidelity. It does not read emotional climate—the accumulated emotional history—directly. But the climate influences the weather through three mechanisms:

1. **Contrast dynamics.** Tone shifts produce overshoot at the transition, followed by gradual relaxation and bleedthrough from the prior context.
2. **Residual traces.** Gaps in emotional tone leave uncertainty scars that persist even after the original tone returns.
3. **Dose-dependent leakage.** Sustained emotional context ( $\geq 8$  turns) eventually leaks to distant token positions, including neutral probes.

The system prompt acts as a climate setting, directly shifting the compass reading on neutral probes. Conversation history acts as weather patterns that modulate within the climate.

### 6.2 Implications for the Oracle Loop

For real-time alignment monitoring, these findings suggest: read the user’s emotional state from the user’s *current* message tokens (high fidelity), not from the model’s response or from distant probe positions. Monitor for contrast dynamics at tone shifts—a sudden shift from positive to negative engagement is *amplified* in the geometry, providing an early warning signal. Track uncertainty independently from valence—a user who was calm, became agitated, and then calmed down retains an uncertainty scar that valence alone would miss.

### 6.3 Architectural Interpretation

The KV cache stores per-position keys that encode each position’s local context. Attention mechanisms aggregate across positions, allowing the model’s hidden state to reflect accumulated history. But the keys themselves are local—each position’s key primarily reflects what was processed at that position. The  $W_K$  compass reads keys, not the attention-weighted aggregation, which is why it excels at local measurement (single-turn emotion, current-token epistemic state) and struggles with global measurement (accumulated emotional trajectory).

This is a feature, not a limitation. The locality of key-based measurement provides a clean,

interpretable signal: the projection at position  $t$  tells you about position  $t$ , without contamination from the rest of the sequence.

## 7 Limitations

1. **Single model.** All experiments use Qwen3.5-27B-Claude-distill.
2. **Small N per cell.** 3 reps per condition; dynamics effects (whiplash  $d=3.6$ , echo  $d=9.9$ ) are large but underpowered. Replication with  $N>30$  per cell is needed.
3. **Composite directions.** Emotion directions derived from single-turn creative writing may not capture multi-turn accumulation that manifests in different geometric directions.
4. **Temperature variation.**  $T=0.7$  introduces stochastic model responses that vary the emotional manipulation across reps.
5. **No behavioral ground truth.** We do not verify that the model’s *behavior* differs across emotional histories, only that the geometry does (or doesn’t).
6. **Position-dependent decay conflated with content.** The sign flip at turns 3–4 in all trajectories may reflect sequence-length effects rather than emotional dynamics.
7. **Variance compression in dose-response.** Pooled SD decreases from 0.52 (dose 1) to 0.31 (dose 4), inflating  $d$  values. Raw mean gaps (0.5–0.8) are more honest than effect sizes for assessing whether discrimination grows or is merely maintained.
8. **Position matching in dynamics comparisons.** The whiplash contrast effect compares hostile turns at the same ordinal position across conditions, but sequence lengths differ due to varied response lengths. Position-matched controls with equalized context length are needed.
9. **No text-feature baseline.** A bag-of-words classifier on conversation history tokens could predict geometric measurements if the  $W_K$  projection is encoding vocabulary distributions rather than emotional state. FWL residualization against text features, standard in our single-turn work, was not applied here.

## 8 Conclusion

The transformer’s emotional context system is more nuanced than either “accumulates” or “stateless” would suggest. The  $W_K$  compass reads current emotional weather with high fidelity and reveals the influence of emotional climate through contrast dynamics, residual traces, and dose-dependent leakage. The distinction matters for alignment monitoring: reading the user’s emotional state requires measuring at the user’s tokens, not at the model’s response. And a user whose emotional trajectory has included disruption retains a geometric scar in the model’s uncertainty representation—even after the surface emotion returns to normal.

The compass reads weather accurately and climate slowly. Both readings are informative. Neither alone tells the whole story.

**Data and code availability.** All experiment scripts and data are available at [Liberation-Labs-THCoalition](#)